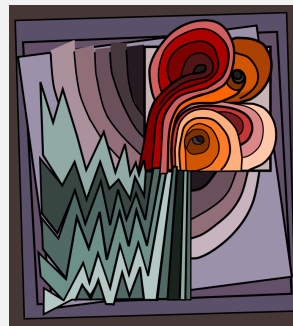


Stochastic observables

A framework for singular fields



UKRI StochFields

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Wilson-Itô programme: joint work with Ismael Bailleul and Ilya Chevyrev

what do we actually observe?

In real life we observe things with finite resolution, we can improve the resolution but never send it to zero.

coarse information $\longrightarrow$ finer information

This process of gathering of more information is encoded in a filtration $(\mathcal{F}_a)_{a \geq 0}$ and all our knowledge in $\mathcal{F}_\infty = \sigma(\cup_a \mathcal{F}_a)$.

For a singular field φ_∞ , the meaningful data may be its observations at every finite resolution $(\varphi_a)_a$

- **Framework.** Consistent observations and equivalence classes of processes.
- **Examples.** Brownian endpoint deltas, the GFF, Wick powers and OPEs.
- **Construction.** Can dynamics in resolution produce an interacting field?

a stochastic observable

Fix a filtration $(\mathcal{F}_a)_{a \geq 0}$. Write $\mathbb{E}_a = \mathbb{E}[\cdot | \mathcal{F}_a]$ and $\mathcal{F}_\infty = \sigma(\cup_a \mathcal{F}_a)$.

Definition. An observable $\mathbb{O} = (O_a)_a$ is a family of integrable, adapted random variables satisfying

$$O_a = \mathbb{E}_a O_b, \quad 0 \leq a \leq b.$$

Its expectation $\langle O \rangle$ is defined as the linear functional $\langle O \rangle = O_0$.

It is a true martingale but no terminal value is required at ∞ .

- ▶ Each O_a is an ordinary random variable.
- ▶ The whole compatible family is the new object.

ordinary random variables sit inside

Every $Y \in L^1(\mathcal{F}_\infty)$ gives

$$O_a = \mathbb{E}_a Y.$$

These are precisely the uniformly integrable martingales (with the usual filtration hypotheses).

$$Y \longleftrightarrow (\mathbb{E}_a Y)_a.$$

Removing uniform integrability enlarges the space of observables.

An almost sure limit alone need not determine the observable.

equivalence classes of descriptions

For integrable adapted processes, set

$$N \sim N' \iff \|N_a - N'_a\|_{L^1} \rightarrow 0 \quad (a \rightarrow \infty).$$

A class contains **at most one martingale** $(O_a)_a$. Indeed, for $a \leq b \rightarrow \infty$,

$$\|O_a - N_a\|_{L^1} \leq \|O_b - N_b\|_{L^1} \rightarrow 0.$$

When it exists, the representative is recovered by looking back:

$$O_a = \lim_{b \rightarrow \infty} \mathbb{E}_a N_b \quad \text{in } L^1.$$

repair a description by its future drift

We will assume all our observation processes are semimartingales and we are in the standard setting (e.g. the filtration is generated by a BM).

Suppose $dN_a = \beta_a da + dM_a$ and

$$\mathbb{E} \int_0^\infty |\beta_c| dc < \infty \text{ or } \mathbb{E} \int_0^\infty |\mathbb{E}_a \beta_c| dc < \infty.$$

Then the observable in its class is

$$O_a = N_a + \mathbb{E}_a \int_a^\infty \beta_c dc.$$

The correction is the expected drift still to come.

$$N_t = B_t + e^{-t} \sin B_t \implies O_t = B_t.$$

Brownian observables

For standard one-dimensional Brownian motion:

$$B_t, \quad B_t^2 - t, \quad B_t^3 - 3tB_t$$

are observables without terminal representatives.

The exponential martingale makes the loss of terminal information explicit:

$$Z_t = \exp [\lambda B_t - \lambda^2 t / 2], \quad \lambda \neq 0,$$

$$Z_t \longrightarrow 0 \text{ a.s.}, \quad \mathbb{E}Z_t = 1.$$

The limiting random variable is zero; the observable is not.

a delta at a Brownian endpoint

Fix $T > 0$ and $y \in \mathbb{R}$. With $p_s(z) = (2\pi s)^{-1/2} e^{-z^2/(2s)}$, define

$$K_t^y = p_{T-t}(B_t - y), \quad t < T.$$

The heat semigroup gives $\mathbb{E}_s K_t^y = K_s^y$. For bounded test functions,

$$\int_{\mathbb{R}} g(y) K_t^y dy = \mathbb{E}_t g(B_T).$$

At a fixed y , $K_t^y \rightarrow 0$ almost surely, but $\mathbb{E} K_t^y = p_T(y) > 0$.

$\delta(B_T - y)$ is represented by its conditional densities.

Here high resolution means $t \uparrow T$; one may reparametrize by $a = (T - t)^{-1}$. Works for Markov processes.

the GFF as an observation process

Let $m > 0$. Couple smooth Gaussian fields X_a by independent increments in resolution, with covariance

$$G_a(x, y) = \int_{a^{-2}}^{\infty} e^{-m^2 s} q_s(x - y) \, ds,$$

$$q_s(z) = (4\pi s)^{-d/2} e^{-|z|^2/(4s)}.$$

Then $X_a(x)$ is a martingale at every point x , it is the GFF $\mathbb{X}(x)$ at x (not a distribution!)

- ▶ $G_a \rightarrow G = (m^2 - \Delta)^{-1}$
- ▶ $v_a = G_a(x, x) \rightarrow \infty$ in $d \geq 2$.
- ▶ For smooth compactly supported h , $X_a(h) \rightarrow X(h)$ in L^2 .

The point observable survives even when the field has only a distributional endpoint.

Local observables of the GFF

Write $v_a = \mathbb{E} X_a^2$.

$$W_a^{(2)} = X_a^2 - v_a, \quad W_a^{(3)} = X_a^3 - 3v_a X_a.$$

The generating function packages every order:

$$\exp [z X_a - z^2 v_a / 2] = \sum_{n \geq 0} \frac{z^n}{n!} W_a^{(n)}.$$

Independent Gaussian increments give

$$\mathbb{E}_a W_b^{(n)} = W_a^{(n)} \quad (b \geq a).$$

Renormalization selects compatible observations, local functions of the field.

more observables than random distributions

At distinct points, the Wick correlation is

$$\langle W^{(n)}(x)W^{(n)}(y) \rangle = n!G(x,y)^n.$$

In $d \geq 3$, $G(x,y) \asymp |x-y|^{2-d}$. The usual L^2 construction after spatial smearing requires

$$n(d-2) < d.$$

- ▶ The Wick square reaches its obstruction at $d=4$.
- ▶ The Wick cube already reaches it at $d=3$.
- ▶ Yet every finite-resolution Wick martingale remains well-defined.

multiplication becomes a partial operation

The pointwise product $O_a P_a$ need not be a martingale.

Define $O \star P$ as the strong asymptotic class of this process. We assume there is a martingale representative.

Assuming $\mathbb{E} \int_0^\infty |d\langle O, P \rangle_c| < \infty$:

$$(O \star P)_a = O_a P_a + \mathbb{E}_a \int_a^\infty d\langle O, P \rangle_c.$$

For the GFF at $x \neq y$,

$$(X(x) \star X(y))_a = X_a(x) X_a(y) + G(x, y) - G_a(x, y).$$

Its expectation is $G(x, y)$. At coincident points, the bracket tail diverges.

Brownian deltas at different targets

For $y \neq z$, the exact product formula is

$$p_\varepsilon(B-y)p_\varepsilon(B-z) = c_\varepsilon e^{-(y-z)^2/(4\varepsilon)} p_{\varepsilon/2}(B - \frac{y+z}{2}).$$

Its L^1 norm tends to zero as $\varepsilon \downarrow 0$. Hence

$$K^y \star K^z = 0 \quad (y \neq z).$$

At equal targets, the exponential suppression disappears and the leading coefficient diverges.

The product depends dramatically on whether the singular supports meet.

Brownian OPE: resolve the delta square

Put $\varepsilon = T - u$, fix $t < u$, and set $c_\varepsilon = (4\pi\varepsilon)^{-1/2}$. Since

$$p_\varepsilon(z)^2 = c_\varepsilon p_{\varepsilon/2}(z),$$

conditioning on the coarse information gives the exact identity

$$\mathbb{E}_t[(K_{T-\varepsilon}^y)^2] = c_\varepsilon p_{T-t-\varepsilon/2}(B_t - y).$$

Use $\partial_s p_s = \frac{1}{2} \partial_y^2 p_s$ to obtain

$$\mathbb{E}_t[(K_{T-\varepsilon}^y)^2] = c_\varepsilon K_t^y - \frac{\varepsilon c_\varepsilon}{4} \partial_y^2 K_t^y + \frac{\varepsilon^2 c_\varepsilon}{32} \partial_y^4 K_t^y + O_{L^1,t}(\varepsilon^{5/2}).$$

the GFF OPE identifies the Wick square

Let $x_{\pm} = x \pm \rho e / 2$, $|e|=1$, $D = e \cdot \nabla$. At fixed resolution a ,

$$(X(x_-) \star X(x_+))_a = G(x_-, x_+) 1 + W_a^{(2)}(x) + \rho^2 C_{2,a}(x) + O_{L^1,a}(\rho^4).$$

Write $\mathcal{W}_a(U, V) = U_a V_a - \mathbb{E}[U_a V_a]$. Then

$$C_{2,a} = \frac{1}{4} \{ \mathcal{W}_a(X, D^2 X) - \mathcal{W}_a(DX, DX) \}.$$

Paulovics: Systematic study of OPE in this framework

local observables

A **local observable** has a *local germ*: a finite-resolution expression which approximates it in L^1 at high resolution.

$$O_a \sim \tilde{O}_a(\varphi_a).$$

For an observable supported in U , the germ uses the field near U on scale a^{-1} .

- ▶ Equivalent germs describe the same observable.
- ▶ Separated regions admit products when the relevant estimates hold.
- ▶ Colliding supports call for OPEs and renormalized local operators.

This points toward a local algebraic description of an euclidean field theory. Closure and locality require proof in each model.

from a language to a construction programme

We would like to specify an interacting field through its local observables.

local germs $\longrightarrow$ compatible martingales $\longrightarrow$ correlations

Wilson-Itô dynamics proposes to generate these objects while progressively revealing spatial information of a “elementary” field coordinate $(\varphi_a)_a$.

Itô: assume that the **scale dynamics** of $(\varphi_a)_a$ is given by an Itô diffusion:

$$d\varphi_a = B_a \frac{da}{a} + dM_a, \quad d\langle M \otimes M \rangle_a = D_a^2 \frac{da}{a}$$

with adapted drift B_a and diffusivity “matrix” D_a^2 . We want that the dynamics is specified only in terms of features of φ_∞ “brought back” to the scale a .

Wilson: assume

- a) the existence of local observables for the “microscopic” drift $(f_a)_{a \geq 0}$ and for the “microscopic” diffusivity $(\Sigma_a^2)_{a \geq 0}$
- b) that the characteristics B_a, D_a^2 of the diffusion at scale a are given by some spatial averaging of the microscopic characteristics:

$$B_a = A_a f_a, \quad D_a^2 = A_a^{1/2} \Sigma_a^2 A_a^{1/2}$$

where $(A_a)_a$ are spatial averaging operators at scale a . E.g.

$$(A_a f)(x) = \int_{\mathbb{R}^d} a^d \chi(a(x-y)) f(y) dy$$

where $\chi: \mathbb{R}^d \rightarrow \mathbb{R}$ is a mass one, positive and positive definite function with support on the unit ball. Note that we could allow also random averaging: $A_a = A_a(\varphi_a)$.

Wilson–Ito diffusions

[I. Bailleul, I. Chevyrev, MG. Wilson–Ito diffusions. arXiv (2023–2026+++)]

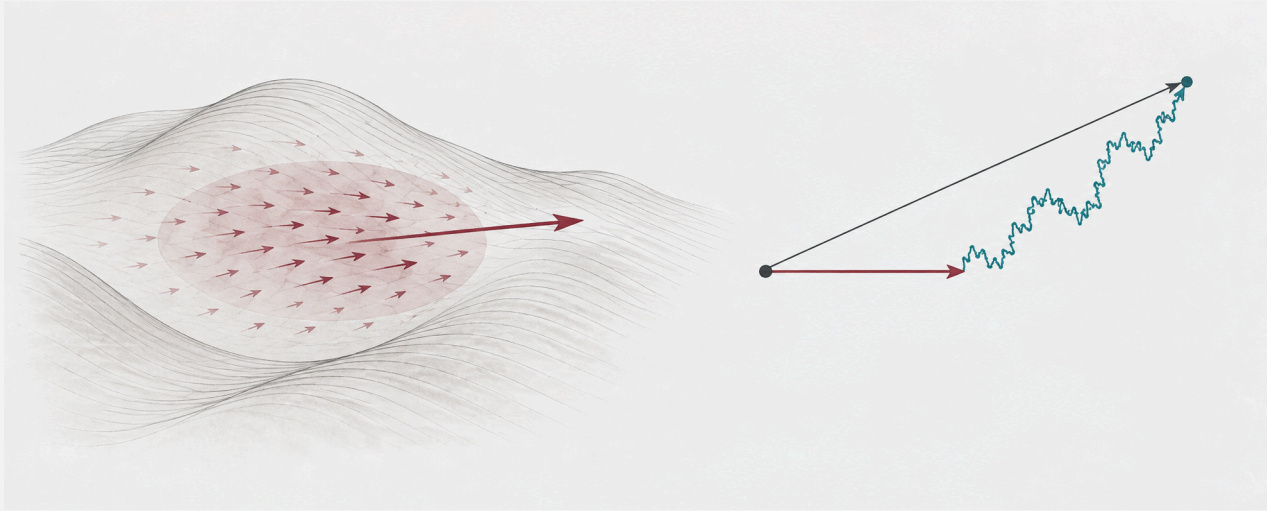
Definition. A *Wilson–Ito diffusion* $(\varphi_a)_a$ is the solution of the SDE

$$d\varphi_a = A_a(f_a) \frac{da}{a} + A_a^{1/2} \sigma_a \frac{dW_a}{\sqrt{a}}. \quad a \geq 0.$$

- ▶ A_a : symmetric positive block averaging, local on scale a^{-1} , tending to **Id**.
- ▶ W : cylindrical Brownian motion.
- ▶ The force f and positive diffusivity σ^2 are local observable fields.

Trivial example · $f_a = 0$, $\sigma_a = 1$. Then $\varphi_\infty = \int_0^\infty A_a^{1/2} \frac{dW_a}{a^{1/2}}$ is a white noise with infinite variance, a reasonable description of the “infinite dimensional” Lebesgue measure.

local averaging, drift and innovation



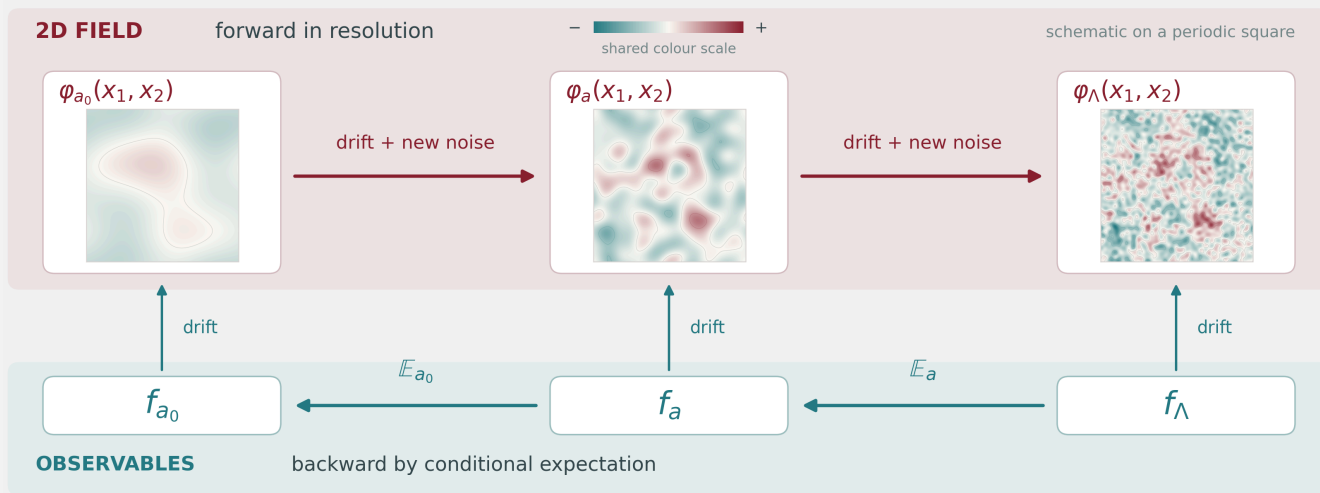
$$d\varphi_a = A_a(f_a) \frac{da}{a} + A_a^{1/2} \sigma_a \frac{dW_a}{\sqrt{a}}.$$

Predictable drift: local forces averaged over a neighbourhood of size a^{-1} .

Unpredictable innovation: fresh noise, with zero conditional mean.

Predictable means known from current information; it need not be deterministic.

a forward field and a backward consistency problem



$$f_a = \mathbb{E}[f_b | \mathcal{F}_a], \quad a \leq b.$$

The field reveals finer spatial information; its local observables constrain the drift and diffusivity that generate it.

are there non-trivial examples?

approximation strategy · fix some $A > 0$ and functional F_A and solve the forward-backward SDE:

$$d\phi_a = A_a \mathbb{E}_a[F_A(\phi_A)] \frac{da}{a} + A_a^{1/2} \frac{dW_a}{a^{1/2}}, \quad a \in (0, A).$$

where $\mathbb{E}_a = \mathbb{E}[\cdot | \mathcal{F}_a]$. Try to send $A \rightarrow \infty$ and at the same time make F_A more and more local.

coherent germs · another approach is to “guess” the drift $f_a \approx F_a(\varphi_a)$ where F_a is the “germ”

$$f_a = F_a(\varphi_a) + R_a$$

then we have a forward–backwards system for (φ_a, R_a) (with $\mathcal{L}_b = \partial_b + \frac{1}{2b} \Delta_{A_b} + \frac{1}{b} F_b A_b \mathbf{D}$)

$$\begin{cases} d\phi_a = A_a (F_a(\varphi_a) + R_a) \frac{da}{a} + A_a^{1/2} \frac{dW_a}{a^{1/2}}, \\ R_a = \mathbb{E}_a \left[\int_a^\infty \mathcal{L}_b F_b(\varphi_b) \frac{db}{b} + \int_a^\infty \mathbf{D} F_b(\varphi_a) A_b R_b \frac{db}{b} \right] \end{cases}$$

Fully open problem in generality · I know very little about it

Semilinear equations

If we assume that the force has the decomposition

$$f_a = -\mathbb{E}_a[L \varphi_A] + \mathbb{E}_a[h(\varphi_A)]$$

define

$$\mathcal{A}_{A,a} = \int_a^A A_b \frac{db}{b}, \quad R_{A,a} = \text{Id} + \mathcal{A}_{A,a} L, \quad \tilde{\varphi}_a^A = R_{A,a}^{-1} \varphi_a,$$

we obtain

$$d \tilde{\varphi}_a^A = C_a \mathbb{E}_a[h(\tilde{\varphi}_A^A)] da + C_a^{1/2} dW_a, \quad C_a := R_{A,a}^{-1} \frac{A_a}{a} R_{A,a}^{-1},$$

The new field $(\tilde{\varphi}_a^A)_a$ has a natural Polchinski-like dynamics, the same terminal value $\varphi_A = \tilde{\varphi}_A^A$ as φ .

Its drift is now only the transformed nonlinear term $C_a \mathbb{E}_a[h(\tilde{\varphi}_A^A)]$, and the linear endpoint force has been fully absorbed into the new operators C_a and $R_{A,a}$.

gradient Wilson–Ito diffusions

Take $h(\psi_\infty) = -DV_\infty(\psi_\infty)$, and let $(V_a)_{a \geq 0}$ be the solution to the *Polchinski equation* (HJB)

$$\partial_a V_a - \frac{1}{2} DV_a \dot{Q}_a DV_a + \frac{1}{2} \dot{Q}_a D^2 V_a = 0$$

then one can prove that $\mathbb{E}_a[h(\psi_\infty)] = -\mathbb{E}_a[DV_\infty(\psi_\infty)] = -DV_a(\psi_a)$ and by performing Doob's h -transform with $d\mathbb{Q} = e^{-V_0(0) + V_a(\psi_a)} d\mathbb{P}$ also that

$$\begin{aligned} \mathbb{E}_{\mathbb{P}}[G(\psi_a)] &= \mathbb{E}_{\mathbb{Q}}[G(\psi_a) e^{V_0(0) - V_a(\psi_a)}] = \mathbb{E}_{\mathbb{P}}[G(X_a^Q) e^{V_0(0) - V_a(X_a^Q)}] \\ &= \frac{\mathbb{E}_{\mathbb{P}}[G(X_a^Q) e^{-V_\infty(X_\infty^Q)}]}{\mathbb{E}_{\mathbb{P}}[e^{-V_\infty(X_\infty^Q)}]} \end{aligned}$$

for any nice function G . In particular the law of ψ_∞ is given by the Gibbs measure

$$\nu_\infty(d\psi) = \frac{e^{-V_\infty(\psi)} \mu^{Q_\infty}(d\psi)}{\int e^{-V_\infty(\psi)} \mu^{Q_\infty}(d\psi)}.$$

Dyson–Schwinger: integration by parts

At a finite-dimensional cutoff, let $\nu(dq) = Z^{-1}e^{-S(q)}dq$, with

$$S(q) = \frac{1}{2}\langle q, Lq \rangle + V(q), \quad L > 0.$$

Integrating $D_h(Ge^{-S})$ gives the Dyson-Schwinger identity:

$$\mathbb{E}_\nu[D_h G] = \mathbb{E}_\nu[G\langle Lq + DV(q), h \rangle].$$

For one coordinate, $S(q) = m^2 q^2 / 2 + \lambda q^4 / 4$ and $G(q) = q^n$:

$$m^2 \mathbb{E}[q^{n+1}] + \lambda \mathbb{E}[q^{n+3}] = n \mathbb{E}[q^{n-1}], \quad n \geq 1.$$

An exact hierarchy: the variance is coupled to the fourth moment.

For $n = 1$: $m^2 \mathbb{E}[q^2] + \lambda \mathbb{E}[q^4] = 1$. Assume sufficient decay and integrability; here $m^2 > 0$, $\lambda \geq 0$.

the Dyson–Schwinger martingale

Put $\mathcal{L}_a = \langle \dot{Q}_a F_a, D \rangle + \frac{1}{2} \text{Tr}(\dot{Q}_a D^2)$ and assume

$$(\partial_a + \mathcal{L}_a)F_a = 0, \quad (\partial_a + \mathcal{L}_a)G_a = 0, \quad DF_a = (DF_a)^*.$$

With $R_a = \dot{Q}_a^{1/2}$, drift cancellation gives

$$d\eta_a = (L - J_a DF_a)R_a dW_a.$$

Both $D_{J_a h}G_a(\varphi_a)$ and $G_a(\varphi_a)\eta_a(h)$ have the same drift:

$$k_a = D_{J_a h}G_a - DG_a \dot{Q}_a DF_a J_a h.$$

Hence the following is a martingale starting at zero:

$$\mathcal{M}_a = D_{J_a h}G_a(\varphi_a) - G_a(\varphi_a)\eta_a(h), \quad \mathbb{E} \mathcal{M}_a = 0.$$

That is, we have

$$\mathbb{E}[D_{J_a h}G_a(\varphi_a)] = \mathbb{E}[G_a(\varphi_a)\langle \eta_a, h \rangle].$$

The equation of motion is already present along the observable flow.

equations of motion as observable identities

For a quartic theory, the nonlinear force becomes a **renormalized insertion**:

$$\mathbb{E}_v[D_h G] = \langle G \langle L\varphi + \lambda[\varphi^3]_{\text{ren}}, h \rangle \rangle_v.$$

Writing $S_2(x, y) = \langle \varphi(x) \varphi(y) \rangle$ gives

$$L_x S_2(x, y) + \lambda \langle [\varphi^3]_{\text{ren}}(x) \varphi(y) \rangle = \delta(x - y).$$

- ▶ The contact term comes from differentiating the observable.
- ▶ The cubic insertion is defined through renormalized correlations; it need not be an ordinary random variable.
- ▶ For the GFF, $\lambda = 0$ and this reduces to $L_x S_2(x, y) = \delta(x - y)$.

Dyson-Schwinger identities constrain the algebra of observables.

Distributional identities, with spatial testing understood; quartic convention $\lambda\varphi^4/4$. Rigorous singular IBP: [Gubinelli–Hofmanova](#); [De Vecchi–Gubinelli–Turra](#).

application: a rigorous sine-Gordon construction

Gubinelli-Meyer: a scale FBSDE constructs the massive sine-Gordon field on $\mathbb{R}^2$ for $\beta^2 < 6\pi$, in the small-coupling regime.

- ▶ A limiting interacting field, with ultraviolet and volume cutoffs removed.
- ▶ Control of local correlations, integrability and large deviations.
- ▶ Singularity relative to the free field in the relevant regime; verification of field-theoretic properties.

For bounded Lipschitz local observables at separated locations:

$$|\text{Cov}_v(O_{1,x}, O_{2,y})| \leq C_{O_1, O_2} e^{-c|x-y|}.$$

The scale estimates give access to correlations, not just to the endpoint law.

The FBSDE approach to sine-Gordon up to 6π (2024, revised 2026).

This is a rigorous scale-diffusion result; it is not a general existence theorem for the current block-average WIDE definition.

application: Φ_3^4 beyond a Gaussian density

Barashkov-Gubinelli: a scale-dependent variational problem constructs the Φ_3^4 measure in finite volume.

- ▶ Renormalized stochastic objects enter the control functional.
- ▶ Coercive estimates and Γ -convergence remove the ultraviolet cutoff.
- ▶ A scale-filtration Girsanov construction gives a random shift of the GFF, followed by an absolutely continuous tilt.

The resulting measure is singular with respect to the unshifted GFF.

The field can be constructed even when a final Gaussian density is unavailable.

Variational method (2020); Girsanov construction (2021). These results support the programme without proving its full local-observable formulation.

Dunlap-Gu: a rigorous forward-backward limit

The two-dimensional nonlinear stochastic heat equation with weak multiplicative noise:

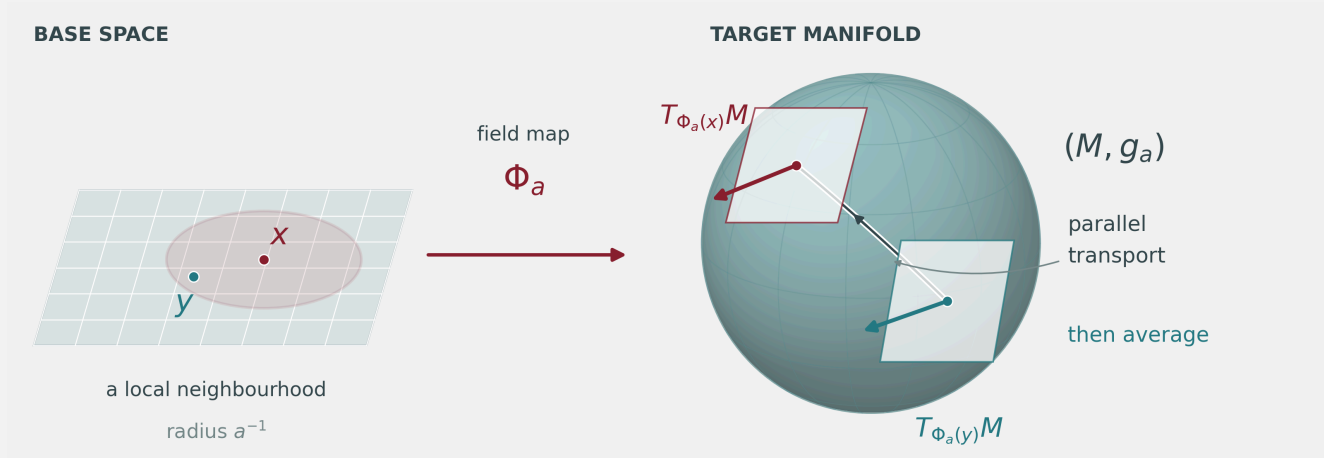
$$\partial_t u_\varepsilon = \frac{1}{2} \Delta u_\varepsilon + \frac{\sigma(u_\varepsilon)}{\sqrt{\log \varepsilon^{-1}}} \zeta_\varepsilon.$$

For constant initial data $x_0 \geq 0$ and $\text{Lip}(\sigma) < \sqrt{2\pi}$, its one-point law at fixed macroscopic time converges to the law of X_2 :

$$dX_q = \sqrt{Y_q} dB_q, \quad Y_q = \frac{1}{4\pi} \mathbb{E}[\sigma^2(X_2) | \mathcal{G}_q], \quad X_0 = x_0.$$

The effective diffusivity is itself a martingale observable.

sigma models: transport before averaging



$$(A_a(\Phi)\omega)(x) = \int K_a(x, y) P_{y \rightarrow x}^{(a)}(\Phi) g_a^{-1}(\Phi(y)) \omega(y) dy.$$

Nearby force covectors live in different fibers. Raise their index and transport them to $T_{\Phi(x)}M$ before adding them.

| sigma models: geometry enters both drift and noise

A proposed intrinsic WIDE, in Stratonovich form, is

$$\delta\Phi_a = A_a(\Phi_a)(f_a)\frac{da}{a} + Q_a(\Phi_a)\frac{\delta W_a}{\sqrt{a}}, \quad \Xi_a = Q_a Q_a^*.$$

For the Dirichlet energy, the bare force is the metric-lowered tension field. On the unit sphere this already contains a nonlinear term:

$$\tau(n) = \Delta n + |\nabla n|^2 n, \quad n(x) \in S^{N-1}, \quad n \cdot \tau(n) = 0.$$

- **Curved target:** tangent noise and parallel transport make the coefficients field dependent.
- **Two base dimensions:** the metric becomes a coupling; its leading perturbative flow has Ricci-tensor form.
- Effective forces require corrections and covariant martingale compatibility.

Geometric WIDE is a programme. Itô coordinates include noise-induced drift. The Ricci term is a perturbative result: [Friedan \(1985\)](#).

gauge fields: the connection determines averaging

For a connection ψ , curvature and the bare Yang–Mills force are nonlinear:

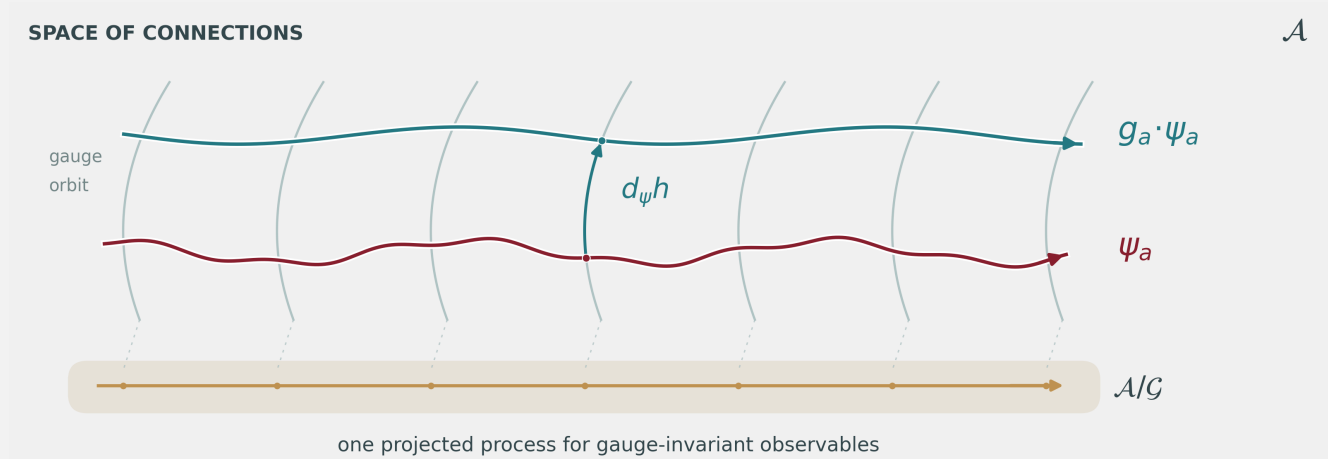
$$F_\psi = d\psi + \frac{1}{2}[\psi \wedge \psi], \quad f^{\text{bare}}(\psi) = -d_\psi^* F_\psi.$$

Let $h^{xy}(\psi)$ transport from y to x . Average Lie-algebra-valued forms in a common frame:

$$(\mathcal{H}_a(\psi)\omega)(x) = \int \chi_a(x, y) \text{Ad}_{h^{xy}(\psi)} \omega(y) dy, \quad A_a = \mathcal{H}_a \mathcal{H}_a^*.$$

- ▶ $h^{xy}(g \cdot \psi) = g(x) h^{xy}(\psi) g(y)^{-1}$ makes the averaging gauge covariant.
- ▶ The field changes the force, the averaging operator and hence the noise covariance.
- ▶ The effective force must satisfy locality and martingale constraints beyond its bare ansatz.

gauge drift changes the representative of the field



$$d\psi_a = (A_a(\psi_a)f_a + d_{\psi_a}h_a)\frac{da}{a} + A_a^{1/2}(\psi_a)\frac{dW_a}{\sqrt{a}}.$$

A regular adapted gauge transformation removes the orbit-tangent term $d_\psi h$, with a rotated Brownian driver.

some remarks on Wilson–Ito diffusions

- ▶ our the working hypothesis is that Wilson–Ito diffusions are natural mechanism to generate and analyse random fields
- ▶ they emerge from simple and natural assumptions and covers in principle much more than those theories that can be reached perturbatively from a Gaussian functional integral, e.g. from the path-integral picture
- ▶ they can be used for gauge theories and fields on manifolds and for Grassmann fields
- ▶ they allow for rigorous non-perturbative results in the whole space
- ▶ (hopefully) they provide a new framework for the stochastic analysis of Euclidean fields
- ▶ numerical simulations?
- ▶ still lot to understand: FBSDEs are non-trivial to analyse but PDE methods seems applicable similarly to Parisi–Wu style stochastic quantisation.

the object is the compatible family

- ▶ **Stochastic observables enlarge random variables.** They retain finite-resolution information without requiring a terminal representative.
- ▶ **Renormalization resolves products into observables.** Brownian deltas and GFF Wick operators show this explicitly.
- ▶ **Wilson-Itô dynamics proposes a construction mechanism.** Forward coordinates and backward consistency must be solved together.

The ambition: construct a field theory together with the observables that make it meaningful.

collaborators



Ismaël Bailleul



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Sarah-Jean Meyer



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Thank you.

The usual order is: a probability law, expectations, then martingales.

Game-theoretic probability reverses this order:

protocol $\longrightarrow$ capital processes $\longrightarrow$ probabilities and expectations.

At round n , a forecast $p_n \in (0, 1)$ and a predictable stake h_n precede the outcome $x_n \in \{0, 1\}$:

$$K_n = K_{n-1} + h_n(x_n - p_n).$$

No probability measure is needed to define this update. Under a compatible law, $p_n = \mathbb{E}[x_n | \mathcal{F}_{n-1}]$, integrable capital processes are martingales.

Nonnegative capital and supermartingale closure are essential. Shafer–Vovk, [Game-Theoretic Foundations for Probability and Finance](#) (2019).

Consider a ticket paying v^0 if $x=0$, and v^1 if $x=1$. Choose

$$k = (1-p)v^0 + pv^1, \quad h = v^1 - v^0.$$

Then the trading strategy exactly replicates the ticket:

$$k + h(x - p) = v^x.$$

- ▶ Backward recursion on a complete finite binary tree recovers ordinary expectation from replication prices.
- ▶ With two fair binary rounds, the ticket paying one for two heads costs $1/4$ initially, then $1/2$ or zero after the first round.
- ▶ The evolving price is the conditional-expectation martingale.

Incomplete protocols lead to upper and lower prices, which need not agree. Shafer, Vovk and Chychyla (2010).

backup / probability bounds are witnessed by processes

The upper probability of an event is the least initial capital needed to cover its indicator, using admissible nonnegative capital processes:

$$\bar{\mathbb{P}}_N(A) = \inf \{k \geq 0: \exists K \in \mathcal{K}_+, K_0 = k, K_N \geq 1_A \text{ on every play}\}.$$

For an admissible nonnegative K with $K_0 = 1$ and $0 < \alpha < 1$, scaling by α and stopping at $1/\alpha$ gives

$$\bar{\mathbb{P}}_N\left(\max_{0 \leq n \leq N} K_n \geq 1/\alpha\right) \leq \alpha.$$

Under a compatible law this is Ville's inequality. A strategy diverging on every violation of a proposed law makes those violations game-theoretically null.

The process is a witness for a probability statement.

Shafer: game-theoretic probability and its uses; test martingales. Infinite-horizon nullity uses the corresponding admissible closures.

Start with continuous price paths S and nonanticipating simple strategies:

$$K_t = k + \sum_i h_i (S_{t \wedge \tau_{i+1}} - S_{t \wedge \tau_i}).$$

- ▶ Suitable pathwise closures define continuous martingales and nonnegative supermartingales, before choosing a probability measure.
- ▶ One then constructs stochastic integrals, quadratic variation and Itô's formula, outside game-theoretically null exceptional sets.
- ▶ Adjoining a derived continuous martingale as a tradable asset does not enlarge the resulting classes: the construction is conservative.

Martingales support the calculus, rather than merely follow from it.

Under compatible measures, these continuous martingales are local martingales; true martingales require integrability. [Vovk–Shafer \(2017\)](#); [martingale spaces](#).

backup / the Brownian delta links the two viewpoints

Normalize the endpoint-delta observable, with $B_0 = 0$:

$$Z_t^y = \frac{p_{T-t}(B_t - y)}{p_T(y)}, \quad Z_0^y = 1, \quad t < T.$$

The backward heat equation gives its capital-process representation:

$$dZ_t^y = \frac{(\partial_z p_{T-t})(B_t - y)}{p_T(y)} dB_t.$$

It is a nonnegative test martingale on every $[0, u]$, $u < T$, yet $Z_t^y \rightarrow 0$ almost surely while $\mathbb{E}Z_t^y = 1$.

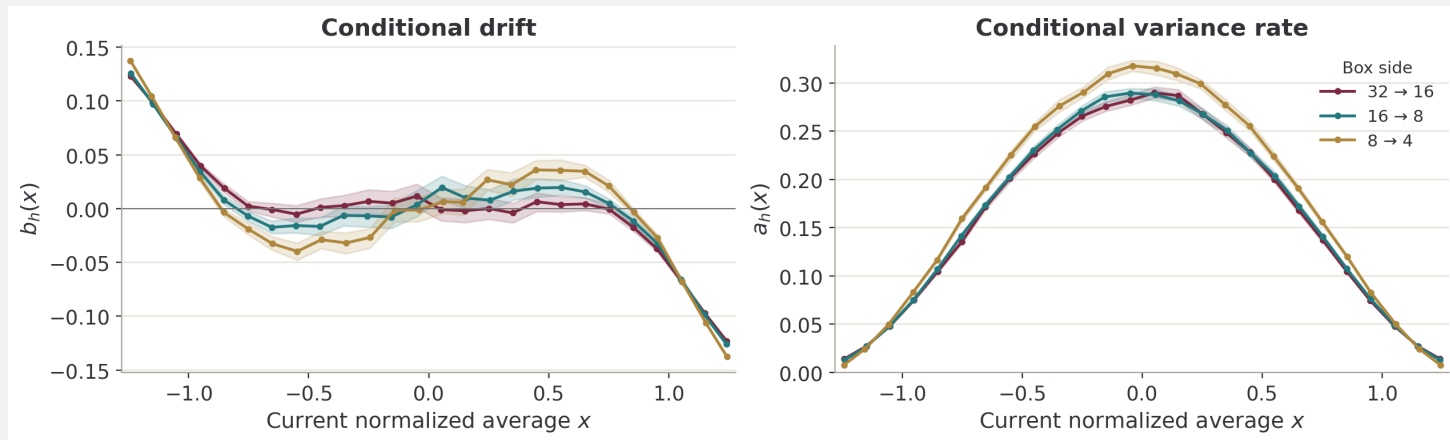
The probabilistic object is the whole martingale, not its terminal limit.

This calculation uses the Brownian law. A pathwise version needs its quadratic-variation clock. Our field framework still uses a fixed law and L^1 ; signed Wick observables need not be nonnegative test capital.

backup / Ising: drift and innovation across resolution

Critical 2D Ising; nested squares; $X_\ell = \ell^{-15/8} \sum_{i \in B_\ell} s_i$, $h = \log(\ell/r)$, $\Delta X = X_r - X_\ell$.

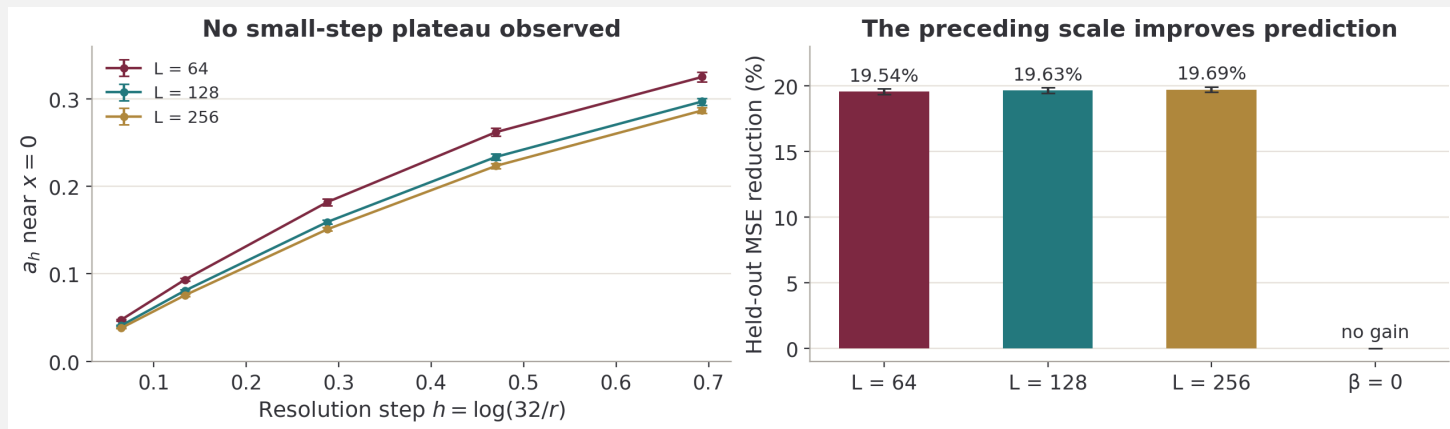
$$b_h(x) = \frac{\mathbb{E}[\Delta X | X_\ell = x]}{h}, \quad a_h(x) = \frac{\text{Var}(\Delta X | X_\ell = x)}{h}.$$



Nonlinear conditional fluctuations; finite-step coefficients. Here a_h is a variance rate, not a noise amplitude.

$L = 256$; 8 chains, 32,768 configurations, 32 centers each. Shading: pointwise 95% block-bootstrap intervals. Equilibrium **Wolff sampling**; resolution is not MCMC time.

backup / Ising: does a scalar diffusion close?



- **Memory:** knowing X_{32} as well as X_{16} improves prediction of $X_8 - X_{16}$ by 19.69% [19.50, 19.90] on held-out chains.
- **Step size:** near zero, a_h falls from 0.2865 for $32 \rightarrow 16$ to 0.0381 for $32 \rightarrow 30$ ($L = 256$).

A single block average is not well closed by these estimates. A field-valued WIDE may need a richer state and a different coupling across scales.

No limiting diffusion coefficient established. Independent-spin control agrees with exact drift and variance; adding history gives no predictive gain.

backup / the correlation hierarchy

Let $S_n(\mathbf{y})$ be the n -point function. Differentiating the product of fields gives contact terms; the force contributes the cubic insertion

$$\mathcal{R}_{n+3}(x; \mathbf{y}) = \left\langle [\varphi^3]_{\text{ren}}(x) \prod_{j=1}^n \varphi(y_j) \right\rangle.$$

The hierarchy is

$$L_x S_{n+1}(x, \mathbf{y}) + \lambda \mathcal{R}_{n+3}(x; \mathbf{y}) = \sum_{j=1}^n \delta(x - y_j) S_{n-1}(\mathbf{\hat{y}}_j).$$

Here $\mathbf{\hat{y}}_j$ omits y_j . The $n+1$ point function couples to a renormalized $n+3$ point insertion.

- ▶ At a cutoff, the cubic term and its counterterms are explicit.
- ▶ In the continuum, take the limit of the whole insertion; diagonal evaluation of a correlation function is not automatic.
- ▶ The identities alone do not provide existence or select an infinite-volume phase.

backup / what this expansion actually says

The OPE resolves the product into a hierarchy of observables:

$$K^y, \quad \partial_y^2 K^y, \quad \partial_y^4 K^y, \quad \dots$$

Their coefficients have successive sizes

$$\varepsilon^{-1/2}, \quad \varepsilon^{1/2}, \quad \varepsilon^{3/2}, \quad \dots$$

The corrections are small at every fixed coarse resolution.

This does not assert that the fine-scale square, minus its leading term, tends to zero in L^1 .

An OPE of conditional observations is weaker than a strong asymptotic product. The delta square has no unrenormalized strong product.

backup / higher operators appear by the same mechanism

At one point, condition a finer-scale polynomial on $\mathcal{F}_a$:

$$\mathbb{E}_a X_b^3 = 3v_b X_a + W_a^{(3)},$$

$$\mathbb{E}_a X_b^4 = 3v_b^2 1 + 6v_b W_a^{(2)} + W_a^{(4)}.$$

Both are exact identities for $b \geq a$.

- ▶ The diverging coefficients multiply lower-order observables.
- ▶ After resolving them, the next renormalized operator is visible.
- ▶ The operator family is selected by compatibility across scales.

Brownian Hermite martingales and GFF Wick fields are instances of the same construction.

backup / an exact finite-cutoff Gaussian calculation

Assume deterministic commuting A_a, L , $L > 0$, $\sigma = 1$, and set

$$f_a = -L\Phi_a^\Lambda, \quad \Phi_a^\Lambda = \mathbb{E}_a \varphi_\Lambda.$$

Define the remaining covariance and its resolvent:

$$H_{\Lambda,a} = \int_a^\Lambda A_c \frac{dc}{c}, \quad R_{\Lambda,a} = \text{Id} + H_{\Lambda,a}L.$$

Conditioning the terminal field gives

$$\varphi_a = R_{\Lambda,a} \Phi_a^\Lambda, \quad d\Phi_a^\Lambda = R_{\Lambda,a}^{-1} A_a^{1/2} \frac{dW_a}{\sqrt{a}}.$$

Exact linear algebra at a fixed cutoff. A compatible continuum WIDE is a further question.

backup / recovering the desired Gaussian covariance

For $\varphi_{a_0}=0$, the previous martingale has covariance

$$Q_a^\Lambda = \int_{a_0}^a R_{\Lambda,c}^{-1} A_c R_{\Lambda,c}^{-1} \frac{dc}{c}.$$

In the commuting case this integrates explicitly:

$$Q_\Lambda^\Lambda = L^{-1} [\text{Id} - (\text{Id} + H_{\Lambda,a_0} L)^{-1}].$$

If H_{Λ,a_0} diverges mode by mode, the endpoint covariance tends to L^{-1} .

Choosing $L = m^2 - \Delta$ recovers the massive GFF covariance.

Endpoint convergence does not prove convergence of the full resolution process or the locality of its limiting observable germs.

backup / coherent germs lead to a forward-backward problem

For a candidate local germ $\tilde{O}_a(\varphi)$, let its drift per $d\log a$ be

$$\mathcal{D}_a \tilde{O} = a \partial_a \tilde{O} + D \tilde{O} [A_a f_a] + \frac{1}{2} \text{Tr}(A_a^{1/2} \sigma_a^2 A_a^{1/2} D^2 \tilde{O}).$$

When the drift tail is integrable and the stochastic term is a true martingale,

$$O_a = \tilde{O}_a(\varphi_a) + \mathbb{E}_a \int_a^\infty (\mathcal{D}_c \tilde{O}_c)(\varphi_c) \frac{dc}{c}.$$

Apply this to the force itself:

$$f_a = \tilde{f}_a(\varphi_a) + r_a, \quad r_a = \mathbb{E}_a \int_a^\infty (\mathcal{D}_c \tilde{f}_c)(\varphi_c) \frac{dc}{c}.$$

Choose and renormalize the germ; estimate the remainder; solve the coupled system.

At a finite cutoff, let X_a be a Gaussian martingale with deterministic covariance C_a , and set

$$h_a(z) = \mathbb{E}[e^{-V_\Lambda(z+X_\Lambda-X_a)}], \quad U_a = -\log h_a.$$

The backward heat equation becomes

$$\partial_a U_a = -\frac{1}{2} \text{Tr}(\dot{C}_a D^2 U_a) + \frac{1}{2} D U_a \dot{C}_a D U_a.$$

The associated Doob transform has dynamics

$$dY_a = -\dot{C}_a D U_a(Y_a) da + \dot{C}_a^{1/2} dB_a.$$

With the corresponding initial law, its endpoint is the Gibbs tilt by e^{-V_Λ} . This covariance parametrization is distinct from the block-average normalization of WIDE.

backup / strong classes versus conditional traces

The existence of $\lim_{b \rightarrow \infty} \mathbb{E}_a N_b$ at each fixed a is weaker than $N \sim O$.

For Brownian motion, take

$$N_t = B_t - B_{(t-1)_+}.$$

Then $\mathbb{E}_a N_b = 0$ for $b \geq a + 1$, but

$$\|N_t\|_1 = \sqrt{2/\pi} \quad (t \geq 1).$$

Its conditional trace is the zero observable, although its strong asymptotic class does not contain zero.

backup / the next GFF OPE coefficient

For the symmetric split and $\mathcal{W}_a(U, V) = U_a V_a - \mathbb{E}[U_a V_a]$,

$$Q_a(\rho) = G(x_-, x_+) 1 + W_a^{(2)} + \rho^2 C_{2,a} + \rho^4 C_{4,a} + O_{L^1,a}(\rho^6),$$

$$C_{4,a} = \frac{1}{192} \{ \mathcal{W}_a(X, D^4 X) - 4 \mathcal{W}_a(DX, D^3 X) + 3 \mathcal{W}_a(D^2 X, D^2 X) \}.$$

Each coefficient is a finite sum of Wick bilinears and is itself an observable.

At fixed a , spatial smoothness gives the Taylor remainder in L^1 .

The remainder constant depends on a . The continuum limit is a separate estimate.

backup / why the delta OPE needs conditioning

At the fine time $T - \varepsilon$, the leading subtraction is

$$(K_{T-\varepsilon}^y)^2 - c_\varepsilon K_{T-\varepsilon}^y = c_\varepsilon [p_{\varepsilon/2}(B_{T-\varepsilon} - y) - p_\varepsilon(B_{T-\varepsilon} - y)].$$

Changing variables $B_{T-\varepsilon} - y = \sqrt{\varepsilon} z$ gives

$$\|(K_{T-\varepsilon}^y)^2 - c_\varepsilon K_{T-\varepsilon}^y\|_1 \sim c_\varepsilon p_T(y) \int_{\mathbb{R}} |p_{1/2}(z) - p_1(z)| dz.$$

This diverges like $\varepsilon^{-1/2}$.

After conditioning at a fixed earlier time, the remainder is instead of order $\varepsilon^{1/2}$.

backup / the finite-cutoff limit is subtle

In a common eigenmode, write $L = \ell > 0$ and $H_{\Lambda,a} = h_{\Lambda,a}$. Then

$$Q_a^\Lambda = \frac{1}{\ell} \left[\frac{1}{1 + \ell h_{\Lambda,a}} - \frac{1}{1 + \ell h_{\Lambda,a_0}} \right].$$

The endpoint covariance tends to ℓ^{-1} as $\Lambda \rightarrow \infty$.

But at a fixed a , both resolvents may tend to zero, so $Q_a^\Lambda \rightarrow 0$.

Correct endpoint laws do not automatically give a nontrivial limiting observation process.

A compatible scale parametrization and limit construction are additional parts of the WIDE problem.

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A. Dunlap, C. Graham, *Renormalization flow for the 2D nonlinear stochastic heat equation: pointwise statistics and universality*, [arXiv:2308.11850](#) (revised 2026).

Model-specific constructions and scaling limits give structural evidence for the local-observable WIDE programme.